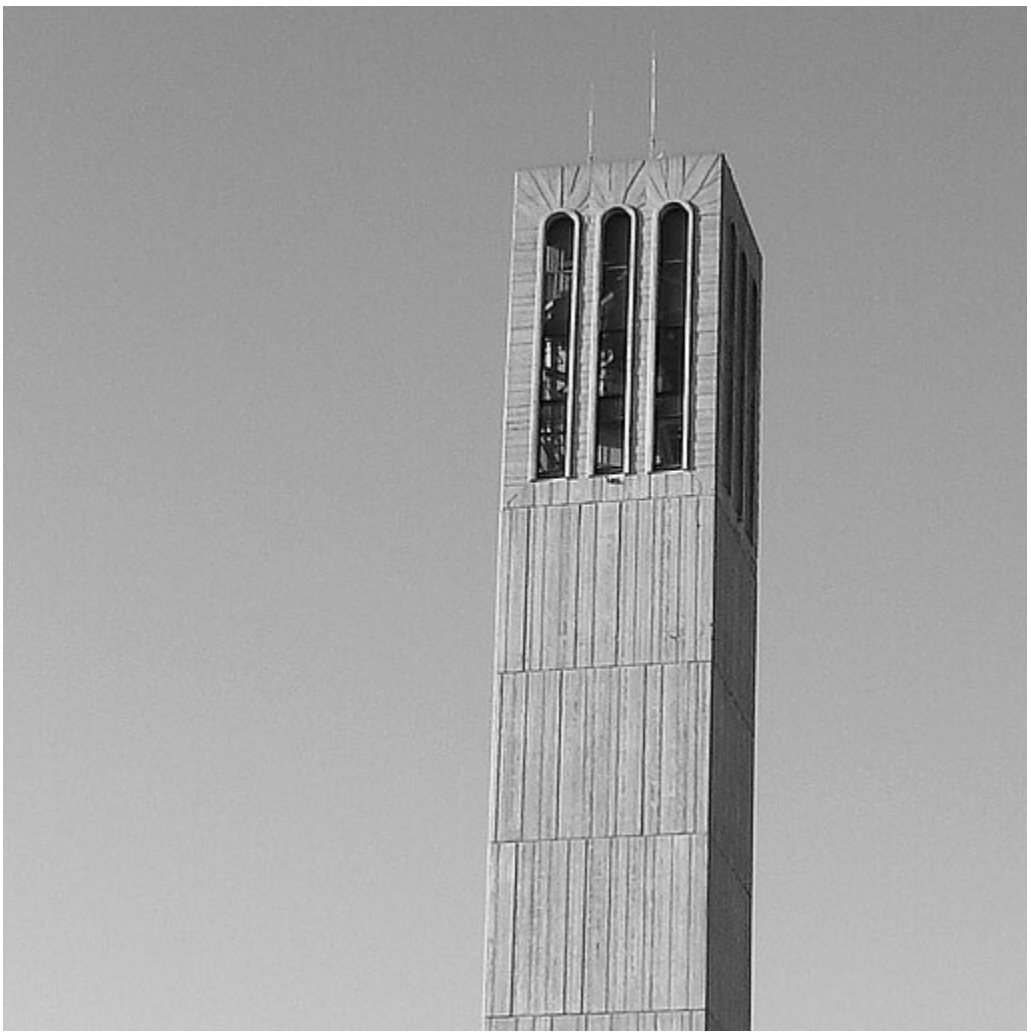
Homework 1 Lab Report

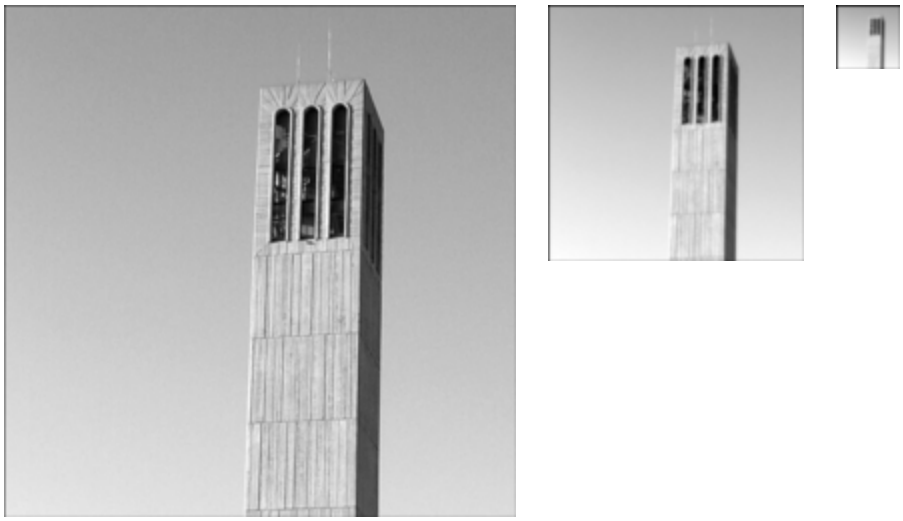
Shravya Salem Sathish

Problem 2a:

Boundary convention: For an image with size (H, W) and a kernel of size (K_H, K_W), full 2D convolution produces an output of size (H + K_H - 1, W + K_W - 1). We want to keep the output image the same size as the input image, so we crop back to the original (H, W) dimensions. However, to keep the output centered, the boundaries are offset by half the kernel dimensions: (K_H - 1) // 2 and (K_W - 1) // 2.

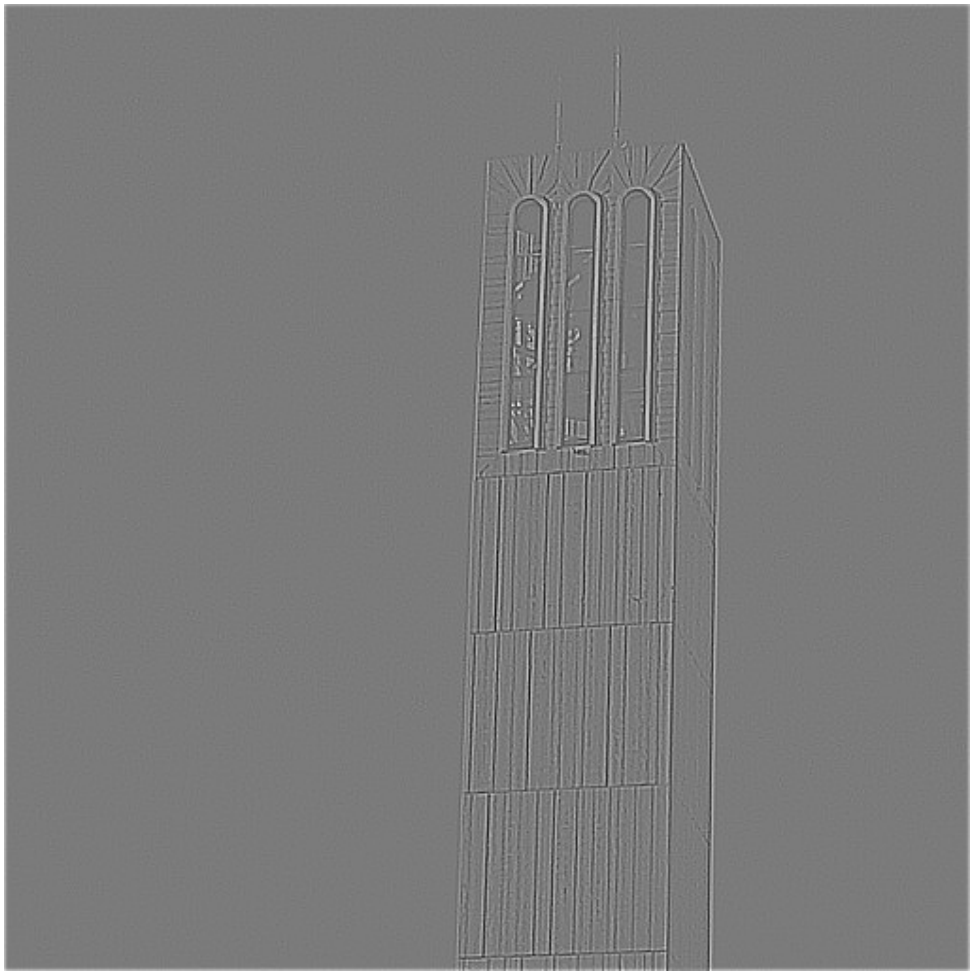
Images: The following four images are the output of the Gaussian pyramid with sizes (512, 512), (256, 256), (128, 128), and (32, 32).

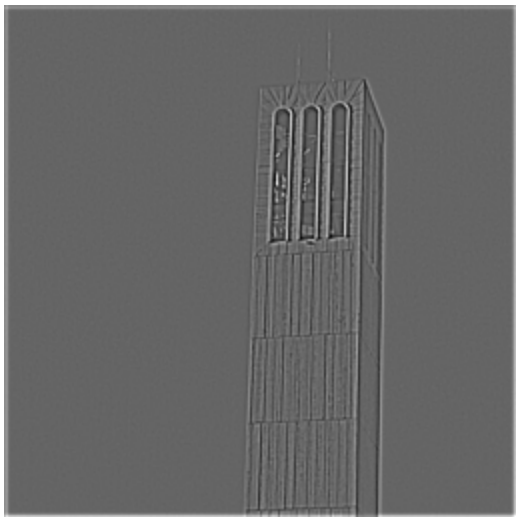
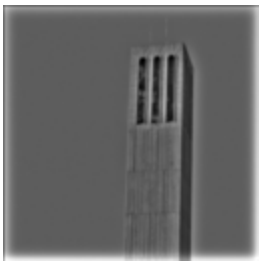




Problem 2b:

Images: The first three images are the output of the Laplacian pyramid with sizes (512, 512), (256, 256), and (128, 128). The fourth image is the coarsest Gaussian residual of size (32, 32)




Problem 3:

Image with detected corners:



Detected corner coordinates:

```
[[ 81  90]
 [ 82  85]
 [ 81  89]
 [144 398]
 [150  72]
 [ 82  89]
 [ 81  85]
 [170 289]
 [151 394]
 [121  11]
 [ 82  90]
 [146 396]
 [169 192]
 [200  92]
 [169 164]
 [141 174]
 [102  85]
 [150  71]
 [ 81  91]
 [201  92]
 [102  84]
 [144 399]
 [101  89]
 [145 396]
 [120  11]
 [167 181]
 [169 191]
 [136 249]
 [139 261]
 [204  92]
 [171 289]
 [144 283]
 [145 288]
 [145 398]
 [ 80  90]]
```

Parameter values:

- `threshold = 35`: Detects 35 candidate pixels with the highest corner response.
- `max_corners = 40`: Detects at most 40 corners. If `threshold > max_corners`, it will detect `max_corners` pixels.
- `k = 0.04`

Results: The pixels with the highest corner response are located at very clear intensity changes. The majority of them are also clustered, so it appears that there are fewer than 35 corners identified. This highlights the importance of non-maximum suppression, which ensures that we locate distinct corners instead of several points that comprise the same corner.

Problem 4a:

I would break down recognizing faces into two steps: (1) detecting that an object is a human face and (2) recognizing the identity associated with it. First, the brain observes basic structure, observing the edges, corners, and contrast of an object to understand its depth and boundaries. These individual pieces are then combined into familiar facial features based on their shape and positions, piecing together an eye, nose, mouth, etc. The overall configuration of these features relative to one another allows us to recognize a whole face due to our learnt knowledge of what faces look like. Processing identity occurs in a different part of the brain, as it requires noticing details idiosyncratic to each person, rather than detecting generic human features. Broadly, I would guess that we compare a perceived face with all familiar faces in our memory and recognize the person if there is a match.

Problem 4b:

I think a computer vision course should focus less on students implementing these pipelines from scratch and more on understanding these systems and evaluating the most efficient solutions. Students should operate more as architects for a problem rather than laborers, making the design decisions on what information needs to be extracted and how features should be detected and classified. It's also still important to have the mathematical and conceptual foundations to explain the algorithms employed by AI, even if they are not writing everything themselves.

Problem 4c:

I'm interested in autonomous vehicles and want to learn how to develop perception systems for them. With my existing experience in sensor and peripheral design, I'd like to integrate my hardware knowledge with what I learn from this course to build systems that can detect obstacles in real time and use that information to control a moving platform. I also want to understand the computer vision algorithms commonly used in industry and how engineers select and adapt them to meet practical constraints.